

Abstract

Understanding and modeling the 3D structure and articulation of everyday objects is a fundamental capability for intelligent robotic systems. Robots tasked with interacting in unstructured environments must not only recognize objects but also reason about their constituent parts and the joints that govern their motion. While recent advances in 3D deep learning have enabled progress in part segmentation and shape reconstruction, the challenge of reliably inferring **joint types, parameters, and articulation models** from partial or noisy observations remains largely unsolved, especially in category-agnostic settings.

This project aims to develop a learning-based framework for **3D per-part segmentation and joint articulation estimation** of articulated objects. Building on recent work in neural implicit representations and category-level articulation modeling, we will train and evaluate models using publicly available datasets such as **PartNet-Mobility** and **Shape2Motion**, which provide fine-grained part annotations and joint labels. The methodology will combine a segmentation backbone (e.g., PointNet++ or transformer-based point cloud encoder) with an articulation prediction module to classify joint types (revolute vs. prismatic), estimate joint parameters (axis, range of motion), and link them to segmented object parts. To address real-world challenges, we will also investigate generalization across object categories and robustness to noisy or incomplete point clouds.

The expected outcomes include:

1. A modular pipeline that performs part-level segmentation and predicts corresponding articulation parameters.
2. Quantitative benchmarks on PartNet-Mobility/Shape2Motion, highlighting strengths and limitations of the proposed approach.
3. A demonstration of the method on simulated interaction scenarios in **SAPIEN**, showing how predicted joints enable object manipulation.

By bridging **3D perception** and **functional understanding of articulation**, this work will contribute toward more capable robotic perception systems and lay the groundwork for future digital twin applications.